Hybrid LiDAR-Supervised Training for Lightweight RGB-Only Monocular Depth in Citrus Orchards: A Branch Study, an Absolute-Scale Fix, and Honest Boundary Limits

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***Abstract*—Lightweight monocular depth estimation from a single RGB camera is attractive for low-cost agricultural robots, but networks developed on urban driving data generalize poorly to vegetation-dense orchards: dense self-similar foliage, shadows, sky with no usable photometric signal, and weak tree–ground boundaries all degrade depth prediction. On CitrusFarm, a citrus-orchard robotics dataset, we find that the standard self-supervised (photometric) recipe is unreliable in this domain—photometric loss can improve while depth metrics and structure get worse—and in particular leaves absolute scale badly wrong (raw abs_rel $= 0.78$, $a_1 = 0.008$ for a plain Citrus fine-tune). We therefore study a *hybrid* regime: keep inference a single RGB image through the unmodified $\sim$3.07M-parameter Lite-Mono depth network, but *train* with projected, densified LiDAR depth and per-pixel valid masks as a masked log-L_1 supervision signal. LiDAR is used only in dataset construction, training supervision, and evaluation—never as an inference input. This single change fixes the scale failure: our 30-epoch hybrid model reaches test raw abs_rel $= 0.1741$ / $a_1 = 0.8186$, versus 0.7787 / 0.0077 for plain Citrus and 0.7273 / 0.0149 for the original baseline. We then run a systematic branch study to test whether additional priors help: a LiDAR-only control (B), an RGB edge-aware smoothness term (C), a hand-coded sky/far prior (F, which *fails* and regresses metrics badly), a frozen original-Lite-Mono prior on LiDAR-invalid pixels (G), and short low-learning-rate retweaks (H/H2). The branch differences are small: most of the gain comes from masked LiDAR continuation, not the extra terms. Our best checkpoint, Branch G `weights_0`, reaches test median abs_rel $= 0.1406$ / $a_1 = 0.8721$ (raw 0.1375 / 0.8739)—but we are explicit that it is the *first* checkpoint of a branch continuing from an already-strong model, that its margin over the LiDAR-only control is within a few thousandths, and that the original prior *drifts* with longer training. We are equally explicit about the limitation that matters for robotics: the model detects rough foreground/object regions but does not draw crisp object outlines. We present this as an honest, reproducible study—a real absolute-scale win for hybrid supervision, modest branch increments, and an unsolved boundary problem—with every number tied to on-disk evaluation artifacts.**

***Index Terms*—monocular depth estimation, lightweight networks, agricultural robotics, LiDAR-supervised training, hybrid supervision, domain shift, negative results.**

I. Introduction

Autonomous agricultural robots—for targeted spraying, yield monitoring, or navigation between tree rows—benefit from dense depth perception that is cheap, lightweight, and camera-only. Lightweight self-supervised monocular depth networks [1], [2] are appealing here: they run from a single RGB image at inference and fit the compute budget of a small robot. However, the canonical training and evaluation domain for these methods is urban driving (KITTI [3]), and vegetation-dense orchards differ sharply: fractal foliage, weak and self-similar texture, wind-induced motion, large bare-ground clearings, and a sky region with no usable photometric signal.

We study this gap concretely on **CitrusFarm** [4], a citrus orchard robotics dataset, using **Lite-Mono** [2] ($\sim$3.07M inference parameters) as the lightweight backbone. Unlike a companion study that constrains training to be strictly label-free, this paper deliberately relaxes the *training* signal while holding the *deployment* interface fixed. Concretely, we keep two constraints: (i) *inference is unchanged*—a single RGB image through the same lightweight Lite-Mono depth decoder, with no architectural addition; and (ii) *LiDAR is training/evaluation-only*—we use projected, densified LiDAR depth and valid masks to supervise training and to score results, but never as an inference input. We call this regime *hybrid*: supervised by LiDAR depth during training, RGB-only at inference.

Our motivation for the hybrid pivot is empirical. We first confirmed that the standard self-supervised recipe is unreliable on CitrusFarm: photometric loss could decrease while Citrus depth metrics and structure got worse, and the resulting models had badly wrong *absolute* scale. A plain Citrus fine-tune reaches only raw abs_rel $= 0.7787$ / $a_1 = 0.0077$ on the test split—essentially no usable metric depth at the predicted scale. The hybrid recipe fixes exactly this failure.

Our contributions are:

1) **A hybrid LiDAR-supervised training recipe** for lightweight orchard depth (Section IV) that fixes absolute scale while keeping inference byte-for-byte identical to the original Lite-Mono. On the CitrusFarm test split it improves raw abs_rel from 0.7787 (plain Citrus) to 0.1741 and raw a_1 from 0.0077 to 0.8186 (Section VI).
2) **A systematic branch study** (Sections V and VI) isolating what actually helps: a LiDAR-only control (B), RGB edge-aware smoothness (C), a hand-coded sky/far prior (F), a frozen original-Lite-Mono prior on LiDAR-invalid pixels (G), and short retweaks (H/H2). The controlled comparison shows that *most* of the gain is masked LiDAR continuation, and that the popular RGB edge-smoothness term is *not* proven useful under this recipe.
3) **An honest account of the best model and its limits** (Sections VI and VIII). Branch G `weights_0` is our best checkpoint, but we show its margin over the LiDAR-only control is within a few thousandths of abs_rel, that it is the *earliest* checkpoint of a branch whose added prior drifts with longer training, that a failed sky-prior branch and two retweak branches did not improve it, and that the model still produces blob-like regions rather than crisp object boundaries—the failure that matters most for a robot.

We are explicit about scope (Section VIII): all results are a single CitrusFarm sequence; the dense LiDAR labels are project-generated, not an official ground-truth product, and cover only $\sim$37% of pixels; and the qualitative depth maps remain weak at object boundaries even where the metrics are strong. We position the work as a careful study of *when* LiDAR supervision buys real gains for lightweight orchard depth and when extra priors do not.

II. Related Work

A. Self-supervised monocular depth estimation

Self-supervised monocular depth turns structure-from-motion into a training signal, removing the need for depth labels. SfMLearner [5] jointly trained a depth network and a pose network from monocular video through a photometric reconstruction loss; Monodepth [6] introduced left–right consistency for the stereo case; and Monodepth2 [1] consolidated the monocular recipe most later work builds on: a combined SSIM $+L_1$ photometric term, a per-pixel minimum over source frames for occlusion, auto-masking of static pixels, and an edge-aware smoothness prior at multiple scales. These components are strong on driving data but rest on the assumption that photometric matching is reliable—an assumption that is fragile in dense vegetation, where repeated leaf texture produces many spurious matches and wind violates the static-scene assumption. Our own experiments (Section **??**) reproduce this fragility: photometric improvement did not translate into depth improvement on CitrusFarm, which is what motivated our hybrid pivot.

B. Lightweight and efficient depth networks

Field robots impose tight compute and memory budgets, motivating architectures that retain accuracy while shrinking parameters and latency. Lite-Mono [2] is a $\sim$3 M-parameter hybrid that interleaves convolutional stages with lightweight self-attention to capture local detail and long-range context at a fraction of the cost of ResNet-based encoders. We adopt Lite-Mono as both backbone and baseline and constrain every contribution to be *training-only*, so the deployed network keeps exactly this footprint and its single-image RGB interface.

C. LiDAR- and depth-supervised monocular training

A long line of work supervises monocular depth networks directly with measured depth. Eigen *et al.* [7] introduced multi-scale supervised regression and the scale-invariant log loss; DORN [8] cast metric depth as ordinal regression; and sparse-depth supervision has been used both alone [9] and as a complement to photometric self-supervision in semi-supervised schemes [10] and in self-supervised scale recovery [11]. Our hybrid recipe is in this family: it supervises a lightweight self-supervised backbone with projected LiDAR depth under a masked log-L_1 objective. The distinguishing constraints in our setting are that (i) the LiDAR labels are project-generated and only $\sim$37% dense, so masking is essential, and (ii) the deployed model must remain RGB-only and lightweight, so LiDAR enters only at training time.

D. Knowledge distillation and frozen-teacher priors

Knowledge distillation transfers knowledge from a teacher into a student by matching the teacher's outputs [12]; in semi-supervised learning, the mean teacher [13] showed that a weight-averaged copy of the student is a remarkably stable teacher. We use a related but deliberately weak idea in one branch (G): a *frozen* original Lite-Mono supplies a normalized-disparity prior only on pixels where LiDAR has no valid label, to preserve broad relative structure outside the supervised region. We report below that this prior helps only marginally and drifts with longer training, so we treat it as a small stabilizer rather than a core mechanism.

E. Depth estimation in agricultural and natural scenes

Most self-supervised depth research is developed and benchmarked on urban driving (e.g. KITTI [3]), whose scene statistics—planar roads, rigid vehicles, strong man-made edges, a clear horizon—differ sharply from an orchard's fractal foliage, weak self-similar texture, non-rigid motion, large bare-ground clearings, and signal-free sky. The CitrusFarm [4] dataset provides synchronized RGB and LiDAR in a citrus orchard and lets us study this domain quantitatively. We treat the orchard not as an incidental test bed but as the central domain that drives both our recipe and our failure analysis.

F. Out of scope: supervised foundation-model distillation

Distilling a large supervised model such as Depth Anything [14] can yield large gains, but imports supervision from the foundation model's labeled pre-training. We do not pursue

it here; our supervision is the project's own LiDAR depth, and we keep the study anchored to a single, auditable label source.

III. Dataset and Evaluation Protocol

Domain and labels. We use CitrusFarm Sequence 01 [4], a citrus orchard sequence with synchronized ZED RGB and Velodyne LiDAR. We build depth labels by projecting LiDAR into the camera frame (route `exact_lidar_parent_child_inverted`) and applying a conservative local inverse-distance-weighted densification (`local_idw`), chosen because it is more conservative in vegetation gaps than broad linear interpolation. Per-pixel valid masks accompany every label; pixels outside the mask are not treated as trusted LiDAR. The matched RGB–LiDAR samples are split into train/val/test by *time blocks* to reduce adjacent-frame leakage. Mean valid-label coverage is ∼37% of pixels (train ∼37.0%, val ∼36.7%). We emphasize that these dense labels are *project-generated*, not an official CitrusFarm ground-truth product.

Integrity statement. The deployed network is RGB-only. LiDAR depth and its valid masks are used in three places: dataset construction, the training loss, and evaluation. They are never an inference input. This is the precise sense in which the method is "hybrid": LiDAR-*supervised* in training, RGB-only at inference.

Metrics. We report standard depth metrics: *raw* (no scale alignment) and *median-scaled* (each prediction divided by the ratio of medians before scoring) abs_rel and the threshold accuracy a_1 ($\delta < 1.25$). Because the central failure of the self-supervised baseline is *absolute scale*, the raw metrics carry the main story here: median scaling hides the very degree of freedom the hybrid recipe fixes. We therefore report both, and lead with raw where scale is the point.

Training recipe. Unless noted, every hybrid model uses: backbone `lite-mono`; batch size 12; 30 epochs; the Lite-Mono learning-rate schedule; masked LiDAR loss of type `log_l1` with weight 0.1; no LiDAR scale alignment (`lidar_scale_align none`); supervision at the full-resolution scale (`lidar_loss_scales 0`); and a minimum of 500 valid supervised pixels per image for the loss to be active. Branch-specific deltas are given in Section V.

Checkpoint selection. Every 30-epoch run is followed by a validation checkpoint sweep and a test confirmation of the top candidates; we do *not* assume the final epoch is best. This protocol is consequential: across runs the selected checkpoint was `weights_13`, `weights_24`, or `weights_0` depending on the run, almost never the final `weights_29`.

IV. Method: Hybrid LiDAR-Supervised Lite-Mono

The core of the method is a single training-only change to the standard Lite-Mono stack: replace the unreliable photometric signal with a masked, LiDAR-supervised metric-depth loss, while leaving the depth network and its inference path untouched.

Masked LiDAR depth supervision. Let $Z_s(p)$ be the student's predicted depth at pixel p, $Z_\ell(p)$ the densified LiDAR label, and $M(p) \in \{0, 1\}$ the LiDAR valid mask. We supervise with a masked log-L_1 penalty,

$$\mathcal{L}_{\text{lidar}} = \frac{\sum_p M(p) \left| \log Z_s(p) - \log Z_\ell(p) \right|}{\sum_p M(p) + \epsilon}, \tag{1}$$

applied with weight 0.1 and *no* scale alignment, so the network is forced to predict depth at the correct *absolute* scale rather than only its shape. The loss is gated by a minimum-valid-pixel count (500) so that images with too little LiDAR coverage do not contribute noisy gradients. Operating in log space balances near and far residuals and matches the orchard's wide depth range.

Why this is the crux. A purely self-supervised monocular objective fixes depth only up to an unknown global scale, and median scaling hides that ambiguity at evaluation. On CitrusFarm this manifests as a catastrophic *raw*-scale failure (Section VI): plain Citrus reaches raw $a_1 = 0.0077$, i.e. almost no pixel is within 25% of the true depth at the predicted scale. Equation (1), supervised directly by metric LiDAR depth with no alignment, is exactly the signal that removes this failure—this is the dominant effect in the paper, far larger than any difference between branches.

What stays fixed. The photometric/reprojection loss is disabled and the pose branch is skipped (it is needed only for photometric warping); only the encoder and depth decoder are loaded and trained. Crucially, the depth network's architecture and parameter count are unchanged, so the inference model remains the original RGB-only Lite-Mono. We measure the deployed model at 3,074,747 inference parameters (2,848,120 encoder + 226,627 depth decoder), ∼11.83 MiB on disk for the encoder+decoder, and ∼15.3–15.6 forward FPS in our GPU evaluator path—squarely within the lightweight Lite-Mono framing.

[FIGURE PLACEHOLDER — insert panel]

Hybrid training vs. RGB-only inference. *Training:* an RGB frame goes through Lite-Mono to a predicted depth; projected/densified LiDAR depth + valid mask supply the masked log-L_1 supervision; the photometric/pose path is disabled. *Inference:* a single RGB image through the unmodified Lite-Mono depth decoder—LiDAR, masks, and all training losses are discarded. Source schematic: to be drawn (see README).

Fig. 1: The hybrid regime. LiDAR depth is used only in dataset construction, training supervision, and evaluation; at inference the model is byte-for-byte the original ∼3.07M-parameter RGB-only Lite-Mono.

V. Branch Study Design

Once the hybrid recipe worked, the open question was whether *additional* priors could improve full-frame visual

TABLE I: Branch study. Each branch changes exactly one term relative to the masked-LiDAR baseline. "From" is the starting checkpoint.

Branch	From	Change vs. masked-LiDAR baseline
A (hybrid)	pretrain	masked LiDAR log-L_1; RGB-only inference (reference)
B	A `w13`	LiDAR-only continuation; `smoothness 0` (control)
C	A `w13`	+ RGB edge-aware smoothness (`smoothness 0.001`)
F	B `w24`	+ hand-coded sky/far + sky-edge contrast priors
G	B `w24`	+ frozen original-Lite-Mono disparity prior on LiDAR-invalid px (`w 0.005`)
H	B `w24`	G prior, 3 ep, LR `1e-5`, `w 0.005`
H2	B `w24`	G prior, 3 ep, LR `1e-5`, `w 0.0025`

sanity and object boundaries without sacrificing the metric-depth win. We answer it with a set of controlled branches, each a full or short training run under the fixed recipe of Section III, differing only in the stated term. All branches except A continue from a common strong hybrid checkpoint, so comparisons isolate the added term rather than confounding it with a different starting point.

B vs. C (the key control). Branch C adds the popular RGB edge-aware disparity smoothness term; Branch B is identical but with the term removed (`disparity_smoothness 0` vs. `0.001`). Because both start from the same checkpoint, dataset, masks, schedule, and seed, the B↔C comparison cleanly tests whether RGB smoothness, rather than simply more LiDAR-supervised training, is responsible for any Branch-C-level gain.

F: a hand-coded sky/far prior. Branch F targets a real visual complaint (sky and upper-canopy regions can look too near, giving a "cave-like" map) with two weak RGB-derived priors: a sky/far prior penalizing near predictions on upper-image sky-colored pixels, and a sky-to-non-sky edge-contrast prior. It is a *repair* fine-tune from the metric-best control, not a fair-from-scratch ablation.

G: a frozen original-prior on unsupervised pixels. Branch G is a safer version of F. Instead of hand-coded rules, it adds a weak normalized-disparity L_1 loss against a *frozen* original Lite-Mono, applied *only* on LiDAR-invalid pixels. Because the target is per-image normalized, it teaches broad relative structure where LiDAR is silent, without disturbing the metric scale that Eq. (1) establishes on valid pixels.

H/H2: stability retweaks. Branch G's best checkpoint turned out to be its *earliest* one, suggesting the prior helps briefly then drifts. H and H2 test whether a short, low-learning-rate application of the same prior (weights 0.005 and 0.0025) reproduces the effect more stably.

TABLE II: The absolute-scale fix, CitrusFarm test split. Hybrid is the 30-epoch masked-LiDAR model (`weights_29`). Lower abs_rel / higher a_1 is better; the dramatic change is in the *raw* columns.

Method	raw abs_rel ↓	raw a_1 ↑	med abs_rel ↓	med a_1 ↑
Original Lite-Mono	0.7273	0.0149	0.3836	0.4989
Plain Citrus (self-sup.)	0.7787	0.0077	0.4889	0.6582
Hybrid 30ep	**0.1741**	**0.8186**	**0.1791**	**0.8001**

VI. Results

A. The absolute-scale fix (the main win)

Table II reports the decisive comparison: the original baseline, a plain Citrus self-supervised fine-tune, and the 30-epoch hybrid model. The hybrid recipe transforms the *raw*-scale metrics—raw abs_rel drops from 0.7787 (plain Citrus) to 0.1741, and raw a_1 rises from 0.0077 to 0.8186 on the test split—while also improving median-scaled metrics. This is the largest effect in the paper and the reason for the hybrid pivot: plain Citrus adaptation produces almost no usable depth at the predicted scale, and direct LiDAR supervision fixes it. Median scaling alone (which both baselines receive) masks this: it grants each prediction a free per-image scale correction, which is why the original baseline's median metrics (0.3836 / 0.4989) look far less broken than its raw metrics (0.7273 / 0.0149).

The hybrid run was also stable, not a lucky final checkpoint: train and validation loss both decreased monotonically in trend (train loss $0.148 \rightarrow 0.072$ across epoch windows; validation $0.154 \rightarrow 0.097$), the median prediction-to-label scale ratio stayed near unity ($\sim$0.985 val, $\sim$1.006 test), and the LiDAR valid fraction held at $\sim$37%. The honest caveat is that most of the gain appeared *early*: the final 30-epoch checkpoint only slightly improved on a 5-epoch checkpoint, so short gates are advisable before committing to long runs.

B. Branch study: most of the gain is LiDAR continuation

Table III reports the branch checkpoints selected by the validation-first protocol. Three findings stand out.

(1) B and C are tied. Branch B (LiDAR-only) and Branch C (LiDAR + RGB edge-aware smoothness) are statistically indistinguishable on every metric (test raw 0.1382 vs. 0.1384; median 0.1436 vs. 0.1437), and their same-image visual panels differ only in tiny, sign-inconsistent local regions. Since the only recipe difference is the smoothness term, *the RGB edge-aware smoothness term is not proven to add value under this recipe*. We report this as useful negative evidence: the gain attributed to a Branch-C-style model is really the gain from masked LiDAR continuation.

(2) The sky prior (F) fails. Branch F regresses badly (test raw abs_rel 0.2492 vs. Branch B's 0.1382). The hand-coded sky/far prior does darken some sky/open regions, but it is too blunt: it damages metric depth and over-fires on non-Citrus vegetation images. We keep it as an informative failure—the idea (full-frame sanity outside the LiDAR mask) is reasonable,

TABLE III: Branch study results, validation and test splits. Selected checkpoints shown. Branch G `weights_0` is the best test abs_rel; the margin over the Branch B control is within a few thousandths. Branch F is the rejected sky-prior branch. Best test values in bold.

		Validation				Test			
Model	ckpt	raw abs_rel ↓	raw a_1 ↑	med abs_rel ↓	med a_1 ↑	raw abs_rel ↓	raw a_1 ↑	med abs_rel ↓	med a_1 ↑
Original Lite-Mono	—	0.7128	0.0195	0.4176	0.4629	0.7273	0.0149	0.3836	0.4989
A / earlier hybrid	`w13`	0.1716	0.8131	0.1746	0.8160	0.1620	0.8149	0.1677	0.8159
B (LiDAR-only)	`w24`	0.1530	0.8672	0.1562	0.8640	0.1382	0.8695	0.1436	0.8661
C (+RGB smooth)	`w24`	0.1531	0.8673	0.1562	0.8641	0.1384	0.8694	0.1437	0.8657
F (sky prior)	`w1`	0.2391	0.8540	—	—	0.2492	0.8564	0.2446	0.8460
G (orig. prior)	`w0`	0.1524	0.8658	0.1564	0.8616	**0.1375**	**0.8739**	**0.1406**	**0.8721**
G (final)	`w29`	0.1582	0.8678	0.1603	0.8635	0.1424	0.8751	0.1456	0.8713
H	`w1`	0.1553	0.8665	—	—	0.1408	0.8733	0.1436	0.8702
H2	`w1`	0.1547	0.8668	—	—	0.1400	0.8739	0.1428	0.8706

[FIGURE PLACEHOLDER — insert panel]

Same-image depth panels (RGB | LiDAR label | Original | Branch B `w24` | Branch G `w0`). Branches B and G are nearly indistinguishable; both fix scale over Original but show soft, blob-like object regions. Source panels: `branch_g/.../checkpoint_sweep/visuals/branch_g_w0_vs_branch_b_w24_test/` and `external_image_sanity_test/outputs_big_original_b_c_f_g_median_scaled/`

Fig. 2: Qualitative comparison. The hybrid branches recover absolute scale and broad layout, but object boundaries remain soft; B and G are visually very close, consistent with their near-identical metrics.

but it needs a stronger (e.g. semantic) mask, not RGB color heuristics.

(3) The original prior (G) helps only marginally, and drifts. Branch G `weights_0` is our best checkpoint (test raw 0.1375 / median 0.1406), but its margin over the Branch B control is within a few thousandths of abs_rel. Critically, `weights_0` is the *first* checkpoint of the branch: it continues from the already-strong Branch B `weights_24` and receives one early, beneficial adjustment from the frozen prior, after which longer training *drifts*—the final `weights_29` is worse on abs_rel (0.1424) despite a marginally higher raw a_1. The retweak branches H/H2 confirm the effect is fragile: neither reproduces or beats `weights_0` (H2, the better of the two, reaches 0.1400), so the improvement is not simply recoverable by lowering the learning rate. We therefore describe Branch G as the best current balance *by the selected metrics*, an incremental win over the LiDAR-only control rather than a new mechanism.

C. Checkpoint selection matters

Across runs, the validation-selected checkpoint was almost never the final epoch: `weights_13` for the earlier hybrid, `weights_24` for B and C, and `weights_0` for G. Treating the final epoch as "the model" would have mis-reported every branch. This is a concrete, reproducible lesson rather than a tuning detail: a validation sweep with test confirmation should be standard for these runs, and the chosen checkpoint depends on the agreed metric priority (e.g. Branch G `weights_29` edges `weights_0` on raw a_1 but loses on abs_rel).

VII. Discussion: the Boundary Failure

The branch study leaves a clear residual problem. Every hybrid model scores well on LiDAR-valid metrics yet, on external sanity images, produces *broad, blob-like* depth regions rather than crisp object outlines. Tree canopies, trunks, flowers, and rocks appear as soft depth blobs; the model knows roughly *where* foreground/background regions are, but not their *shape*. This is visible even for Branch G, our best checkpoint.

We attribute this to a structural mismatch between the supervision and the goal. The LiDAR-valid mask covers only ∼37% of pixels and is dominated by ground/structure surfaces that return clean LiDAR; foliage boundaries, thin structures, and sky are exactly where the labels are sparsest or absent. A model can thus minimize masked abs_rel while remaining blind to object boundaries, because the metric never strongly penalizes boundary error. Branch F and Branch G were both attempts to inject a signal *outside* the LiDAR mask—F by hand-coded sky rules (too blunt), G by a frozen relative-disparity prior (too weak to sharpen boundaries). Neither produced object-aware structure. The lesson is that generic, un-targeted priors do not teach boundaries; an explicit, trustworthy object/semantic signal is likely required.

This matters for the deployment motivation. A spraying or navigation robot needs reliable plant/obstacle *structure*, not only low average LiDAR-valid depth error. We therefore explicitly decline to claim that the method solves object-boundary depth; it improves metric depth and absolute scale, and leaves boundary sharpness as the principal open problem.

VIII. Limitations

We are explicit about what the results do *not* establish.

Branch increments are small. The headline branch result (G over B) is within a few thousandths of abs_rel, the best checkpoint is the branch's earliest one, and the added prior drifts with training. We report G as the best balance by the

selected metrics, not as a robust new mechanism; the proven, large effect is the LiDAR supervision itself, not the priors layered on top.

Metrics improve, appearance does not. Hybrid training fixes scale and lowers error, but object boundaries remain soft (Section VII); the qualitative depth maps are not visibly transformed.

Project-generated labels. The evaluation depth is derived from LiDAR by our own projection and densification, not an official ground-truth product, and covers only $\sim$37% of pixels. Strong masked metrics therefore certify ground/structure depth far more than foliage-boundary depth.

Single domain. All results are CitrusFarm Sequence 01. Generalization to other orchards, crops, seasons, or lighting is unverified; external-image inspections are qualitative sanity checks, not quantitative benchmarks.

Negative branches are not exhaustive ablations. F, G, and H/H2 are targeted experiments from the metric-best control rather than fair-from-scratch ablations, so they bound *these implementations* of each idea, not the idea in general (e.g. a semantic-masked sky prior might still help).

IX. Future Work

Directions implied by our analysis: (i) semantic- or segmentation-assisted object and sky masks during training, to inject a *trustworthy* boundary signal where LiDAR is silent—the targeted version of what F and G attempted bluntly; (ii) edge-aware losses gated by reliable object/semantic edges rather than generic RGB gradients; (iii) better LiDAR label-confidence and interpolation-uncertainty handling, so training does not over-trust densified pixels; (iv) a controlled early-stop protocol around the Branch G first-epoch behavior, since the useful effect is concentrated there; and (v) cross-sequence and cross-orchard generalization. Foundation-model distillation [14] is a separate, higher-supervision path we leave out of this LiDAR-only study.

X. Conclusion

We presented a hybrid, inference-unchanged training recipe—masked LiDAR-supervised log-L_1 depth on a lightweight RGB-only Lite-Mono—that fixes the absolute-scale failure of self-supervised adaptation in a citrus orchard, improving test raw abs_rel from 0.7787 to 0.1741 and raw a_1 from 0.0077 to 0.8186 while keeping the deployed model byte-for-byte identical to the original $\sim$3.07M-parameter network. A systematic branch study then showed, honestly, that most of the gain is the LiDAR supervision itself: an RGB edge-smoothness term ties the LiDAR-only control, a hand-coded sky prior fails, and a frozen original-prior yields only a marginal, drift-prone increment whose best checkpoint (Branch G `weights_0`, test median abs_rel $= 0.1406$ / $a_1 = 0.8721$) is the branch's earliest. We are equally clear about the limitation that matters for robotics: the model detects rough regions but not crisp object boundaries. We hope the combination of a real absolute-scale win, carefully bounded branch increments, and a transparent boundary-failure analysis is useful for future lightweight depth work in agricultural domains.

Reproducibility and Honesty Notes

All reported numbers are taken from on-disk evaluation artifacts (validation/test summary JSONs and per-sample CSVs) produced by a fixed evaluator; the hero Branch G `weights_0` result was re-confirmed by a clean checkpoint resweep. Training is LiDAR-supervised; inference is single-image RGB-only Lite-Mono with no architectural change. Figures 1–2 are placeholders whose on-disk source panels are named in the captions and README; several references are marked unverified in the bibliography and must be checked before submission.

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